

Pump P-101 Standard Operating Procedure (SOP)

1. Document Information

Equipment Tag: P-101

Equipment Type: Centrifugal Process Pump

Procedure: Inspection, Condition Monitoring and Troubleshooting

Document Status: Synthetic Demonstration Document

Revision: 1.0

Note: This SOP contains synthetic procedural information created for the Sovereign AI Workbench demonstration. It does not represent an actual industrial site's operating procedure.

2. Purpose

This procedure defines the general inspection, condition-monitoring, troubleshooting, and response process for Pump P-101.

The objective is to identify abnormal operating conditions early, protect equipment from damage, and ensure that appropriate personnel are notified when intervention is required.

3. Scope

This procedure applies to routine monitoring and troubleshooting of Pump P-101.

The procedure covers:

- Vibration monitoring
 - Temperature monitoring
 - Pressure monitoring
 - Visual inspection
 - High-vibration response
 - Basic troubleshooting
 - Controlled shutdown considerations
 - Maintenance escalation
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4. Safety Requirements

Before performing any physical inspection or maintenance:

1. Follow applicable site safety procedures.
2. Use appropriate personal protective equipment.
3. Keep clear of rotating equipment.

4. Do not touch moving components.
 5. Follow isolation and lockout/tagout procedures when required.
 6. Confirm that equipment is in a safe condition before inspection.
 7. Do not bypass safety interlocks or protective devices.
 8. Contact responsible maintenance personnel when the condition exceeds the operator's authority.
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5. Normal Operating Parameters

Parameter	Normal Range
Speed	1450 RPM
Flow	Approximately 120 m ³ /h
Discharge Pressure	4.0–4.5 bar
Temperature	60–75 °C
Vibration	Below 4.5 mm/s RMS

These values are synthetic demonstration limits and should not be treated as limits for real industrial equipment.

6. Vibration Classification

Vibration	Condition	Response
< 4.5 mm/s RMS	Normal	Continue monitoring
4.5–7.1 mm/s RMS	Warning	Investigate condition
> 7.1 mm/s RMS	Critical	Immediate investigation and controlled response

A single sensor value should be verified before major maintenance action is taken.

Trend information should also be reviewed.

7. Routine Monitoring Procedure

During normal operation:

1. Record vibration.
2. Record bearing or equipment temperature.
3. Record discharge pressure.

4. Record flow where available.
5. Record motor speed.
6. Compare readings with normal operating limits.
7. Review trends for gradual deterioration.
8. Record abnormal observations.

Any sustained deviation should be investigated.

8. High-Vibration Response Procedure

When vibration exceeds the normal operating range:

Step 1 — Verify the Reading

Check the sensor reading and confirm that the measurement is valid.

Step 2 — Review Other Parameters

Check:

- Temperature
- Pressure
- Flow
- Speed
- Recent operating changes

Step 3 — Review History

Check recent:

- Maintenance activities
- Previous vibration events
- Bearing work
- Coupling alignment work
- Equipment changes

Step 4 — Inspect

When safe and authorized, inspect for:

- Bearing abnormalities
- Coupling misalignment
- Mechanical looseness
- Seal problems

- Unusual noise
- Visible damage

Step 5 — Escalate

Notify the responsible maintenance or engineering personnel if the condition persists or worsens.

9. Critical Vibration Response

If vibration exceeds **7.1 mm/s RMS**:

1. Verify the sensor reading.
 2. Notify the responsible operator or maintenance personnel.
 3. Review temperature, pressure, flow, and speed.
 4. Avoid unnecessary continued operation.
 5. Follow the applicable controlled-shutdown procedure when required.
 6. Inspect bearing and coupling condition.
 7. Investigate possible mechanical causes.
 8. Document the event.
 9. Do not return the equipment to normal service until the responsible personnel determine that it is safe.
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10. Troubleshooting Guide

Observation	Possible Cause	Recommended Investigation
High vibration	Bearing degradation	Inspect bearing
High vibration	Coupling misalignment	Check alignment
High vibration	Mechanical looseness	Inspect mounting and fasteners
High temperature	Bearing/lubrication problem	Inspect bearing and lubrication
Low pressure	Flow/process problem	Check operating conditions
Abnormal noise	Mechanical problem	Inspect pump and rotating components
Seal leakage	Mechanical seal issue	Inspect seal

These possible causes are not definitive diagnoses. Physical inspection and supporting evidence are required.

11. Restart Procedure

After maintenance:

1. Confirm maintenance activities are complete.
 2. Confirm tools and temporary equipment have been removed.
 3. Confirm guards and protective components are correctly installed.
 4. Confirm the equipment is ready for operation.
 5. Start the pump according to the applicable operating procedure.
 6. Monitor vibration and temperature.
 7. Verify pressure and flow.
 8. Record post-maintenance readings.
 9. Report any continuing abnormal condition.
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12. Documentation Requirements

The following information should be recorded for abnormal events:

- Date and time
 - Equipment ID
 - Sensor readings
 - Observed symptoms
 - Operator observations
 - Investigation findings
 - Maintenance performed
 - Parts replaced
 - Post-maintenance readings
 - Final equipment condition
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13. Escalation

Escalate the condition to responsible engineering or maintenance personnel when:

- Vibration exceeds the critical limit.
- Vibration continues to increase.
- Temperature becomes abnormal.
- Mechanical damage is suspected.
- Repeated alarms occur.

- Equipment protection devices activate.
 - Safe operation cannot be confirmed.
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14. Relationship to AI-Based Monitoring

The Sovereign AI Workbench may identify abnormal sensor behavior using data analytics.

For example:

Vibration = 8.7 mm/s RMS

The analytics system may identify this as a high-vibration anomaly.

This SOP can then be retrieved by the RAG system to provide the appropriate investigation and response procedure.

The SOP does not replace qualified personnel, site procedures, or safety requirements.

15. Document Control

Equipment: P-101

Document: Standard Operating Procedure

Revision: 1.0

Purpose: Synthetic demonstration and AI/RAG testing

Document Type: Demonstration data only

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